Mathematics for Computing - ICT 2 Practice Questions

Mathematics for Computing (4COSC007C) · December 2023 · typed up from tutorial, mock paper and class photos

About ICT 2 (In-Class Test, Written Examination)

- **Date** / **location:** exam schedule announced later; be present on campus at the given time.
- **Mode:** written exam with an exam paper and an official answer booklet. You may make notes on the exam paper, but only answers in the answer sheet are graded.
- **Format:** closed book; no textbooks, notes or online resources. Each question carries different marks.
- **Duration:** 90 minutes. Missing your assigned timeslot gives a zero mark.
- **Pass criteria:** a minimum mark of 30 in each of ICT1 and ICT2, and a combined overall mark across ICT1 and ICT2 of at least 40.
- **Bring:** student ID, a pen and a calculator. Blank paper for notes is not permitted.
- **Revise:** Lecture Notes 5-10 - Graph Theory, Matrices, Probability Theory, Statistics Basics. A Mock Exam Paper is on Blackboard (In Class Tests > ICT2 Mock Test).

Matrices

Question 3.3 (d, e)

d) Given A = $(1, 3)^T$ and B = $(2, 3)^T$ (column vectors), multiply A^T and B.

e) Given A = B = [[2, -3], [3, -2]], multiply A^T and B.

Question 3.4

Let A be an identity matrix of size 5. Find: a) A^2 b) A^{10} c) A^{100} d) A^T e) A^{-1}

Question 3.5

For which of the following matrices can we compute A^2 and A^3? If the multiplication is possible, compute the result; otherwise explain why it is not possible.

1. A is of size 4x5
2. A is a square matrix of size 2 with all elements 2
3. A is a row vector with 3 columns
4. A is a column vector with 6 rows
5. A is a diagonal matrix whose main diagonal has 3 elements: 1, 2 and 3
6. A is a null matrix of size 2x3

Question A3 (7 marks)

You are given the following equation, where X is an unknown matrix:

$$3X - \begin{vmatrix} 2 & 5 \\ 3 & 6 \end{vmatrix} = 7X + \begin{vmatrix} 4 & -3 \\ -2 & 1 \end{vmatrix}$$

Find the values of X_{11} and X_{22} in the matrix X that satisfy this equation.

Graph Theory

Question Graph N

Graph N has nodes 1, 2, 3 and 5, with directed edges 1→2, 1→3, 2→3 and a self-loop on node 5 (node 5 is not connected to the others).

1. Form the graph notation for N: a set V of all its nodes and a set E of all its edges.
2. Determine if it is a directed or undirected graph.
3. Determine if it is a connected or disconnected graph.
4. Determine if it is a cyclic or acyclic graph.

Question Task 2 - Weighted graphs

You are part of a network optimisation team analysing data packet transmission from a central server in London to data centres across the UK. Construct a weighted directed graph of the network latency from the central server to each data centre, using the data recorded on Monday 30 October 2023 between 8 am and 10 am:

- Manchester: 50 ms at 8 am
- Birmingham: 35 ms at 9:30 am
- Bristol: 45 ms at 9:30 am
- Southampton: 40 ms at 8:30 am
- Glasgow: 65 ms at 8 am
- Edinburgh: 70 ms at 8 am

1. **Define the set of nodes V:** one node per data centre (not the London server, which is the origin).
2. **Construct the set of edges E:** for each pair x, y in V, create a directed edge (x, y, w) if and only if the latency from London to x is less than the latency to y. The weight w is the latency to y minus the latency to x.
3. **Graph creation:** draw the weighted directed graph from V and E.
4. **Analysis:** briefly analyse the graph (distribution of latencies, implications for routes, patterns that might influence network optimisation).

Question Task 4 - Critical path analysis

The project schedule below is a network graph: each node is a task and each edge is an activity with its duration. Find the earliest and latest times for each node and the critical path.

Activity	From	To	Duration
A	1 (Start)	3	11
B	1 (Start)	2	33
K	2	3	1
C	2	5	154
I	3	4	22
J	4	7	143
D	5	7	22
H	5	6	11
E	6	7	1
G	7	8 (Finish)	11

Statistics (ICT2 Mock Exam Paper 2023)

Question RTT

Round-trip times (RTT) of six packets:

Packet	P1	P2	P3	P4	P5	P6

RTT (ms)	45	60	45	75	60	60

Calculate: A. Mean (average RTT) B. Range C. Median D. Mode E. Standard deviation (RTT consistency) F. Variance (degree of RTT spread) G. Interquartile range (spread of the middle 50% of RTTs)

Answers and Worked Solutions

3.3 d) A^TB = 1x2 + 3x3 = **11** (a 1x1 matrix).
3.3 e) A^T = [[2, 3], [-3, -2]]; A^TB = [[13, -12], [-12, 13]].

3.4 For the identity matrix I_5, all five are I_5: $A^2 = A^{10} = A^{100} = A^T = A^{-1} = I_5$.

3.5 A^2 exists only for square matrices.

- i. 4x5: not square, so not possible (4x5 times 4x5 is undefined).
- ii. [[2,2],[2,2]]: A^2 = [[8,8],[8,8]], A^3 = [[32,32],[32,32]].
- iii. 1x3 row vector: not possible.
- iv. 6x1 column vector: not possible.
- v. diag(1,2,3): A^2 = diag(1,4,9), A^3 = diag(1,8,27).
- vi. 2x3 null matrix: not square, so not possible.

A3 3X - M = 7X + N, so -4X = M + N = [[6, 2], [1, 7]] and X = [[-1.5, -0.5], [-0.25, -1.75]]. Therefore **X_{11} = -1.5** and **X_{22} = -1.75**.

Graph N (solution given in class): A. V = {1, 2, 3, 5}, E = {(1,2), (1,3), (2,3), (5,5)} B. Directed C. Disconnected D. Cyclic (the self-loop on 5 is a cycle).

Task 2 V = {Manchester, Birmingham, Bristol, Southampton, Glasgow, Edinburgh}. E has 15 edges, one from each lower-latency centre to each higher-latency centre:

From (x)	To (y)	w (ms)
Birmingham	Southampton	5
Birmingham	Bristol	10
Birmingham	Manchester	15
Birmingham	Glasgow	30
Birmingham	Edinburgh	35
Southampton	Bristol	5
Southampton	Manchester	10
Southampton	Glasgow	25
Southampton	Edinburgh	30
Bristol	Manchester	5
Bristol	Glasgow	20
Bristol	Edinburgh	25
Manchester	Glasgow	15
Manchester	Edinburgh	20
Glasgow	Edinburgh	5

Analysis: latencies range from 35 ms (Birmingham) to 70 ms (Edinburgh). The Scottish centres (Glasgow, Edinburgh) are 15-35 ms slower than the English ones, so they are the first candidates for route optimisation or a closer relay. Measurements were taken at different times, so time of day may also affect the results.

Task 4 (solution worked on the whiteboard in class)

Node	1	2	3	4	5	6	7	8
Earliest	0	33	34	56	187	198	209	220
Latest	0	33	44	66	187	208	209	220

Critical path: **1 → 2 → 5 → 7 → 8** (B, C, D, G) with total duration **220**. These are the nodes whose earliest and latest times are equal.

RTT Sorted data: 45, 45, 60, 60, 60, 75 (n = 6).

- A. Mean = 345 / 6 = **57.5 ms**
- B. Range = 75 - 45 = **30 ms**
- C. Median = (60 + 60) / 2 = **60 ms**
- D. Mode = **60 ms**
- F. Variance = sum of squares / (n - 1). Squared differences: 156.25, 6.25, 156.25, 306.25, 6.25, 6.25; sum = 637.5; variance = 637.5 / 5 = **127.5 ms^2**
- E. Standard deviation = √127.5 ≈ **11.29 ms**
- G. Q1 = ¼(n+1) = 1.75th item = 45; Q3 = ¾(n+1) = 5.25th item = 60 + 0.25 x 15 = 63.75; IQR = **18.75 ms**

The mock solution in the photo used 272.25 for (75 - 57.5)^2 (underlined in the photo), giving a sum of 603.25, variance 120.65 and SD 10.99. The correct value is 306.25, giving the figures above. The official solution file is in this folder as ICT2-Solutions-MockExamPaper-2023.docx.